

LASSO-Based Feature Selection for High-Dimensional Genomic Data

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1. Project Significance

- Genomic data typically contain tens of thousands of genes (variables), while patient samples number only in the tens to hundreds, creating a high-dimensional ($n < p$) setting where variables far exceed samples. Under these conditions, ordinary regression tends to overfit by learning noise, and the estimated coefficients become unstable, changing drastically with small shifts in the data.
- Among tens of thousands of genes, only a small fraction are truly related to disease, and identifying them accurately carries clinical importance since they can become biomarkers for early diagnosis or targets for new treatments.
- LASSO is a leading method for addressing this high-dimensional problem, but it has several limitations : it often fails to jointly select correlated genes, produces unstable results, and can't directly test statistical significance. Methods such as Elastic Net, Adaptive LASSO, Random LASSO, Hi-LASSO, Stochastic LASSO have been proposed to address these gaps, yet limitations remain when applied to real, high-dimensional genomic data with strong correlation structures. This project plans to apply these methods to both simulated and real data and compare their performance.

2. Literature Survey

2-1) Random LASSO

Random LASSO is a bootstrap-based method designed to address the instability of variable selection in standard LASSO. It repeatedly draws random subsets of features, applies LASSO to each subset to estimate variable importance, and then resamples important variables more frequently in a second round of selection. This approach allows

correlated variables to be selected together, permits selection of more variables than samples, and improves reproducibility through repeated averaging.

2-2) Hi-LASSO

Hi-LASSO improves on Random LASSO by statistically correcting the bias introduced during bootstrap resampling and by testing the statistical significance of each variable based on its selection frequency. It also determines the number of bootstrap iterations statistically, rather than relying on an arbitrary threshold as in earlier methods. As a result, Hi-LASSO addresses the significance-testing gap left open by Random LASSO and increases the reliability of variable selection.

2-3) Stochastic LASSO

Stochastic LASSO further improves on Hi-LASSO by addressing multicollinearity that still remains within individual bootstrap samples and by correcting uneven feature coverage across resampling. It constructs bootstrap subsets to avoid drawing highly correlated variables together, ensures more balanced sampling across all features, and applies a two-stage t-test for significance testing. On real GBM genomic data, Stochastic LASSO achieved an F1 score of 0.51, whereas Hi-LASSO failed to detect any significant variables.

3. Methods

3-1) Simulation Data

To validate feature selection performance, we generated simulation data with known ground truth. Following the settings commonly used in existing bootstrap-based LASSO studies, we generated four synthetic datasets (Dataset I–IV) using the following linear regression model:

$$y = \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_p x_p + \epsilon,$$

where $\epsilon \sim N(0, \sigma^2)$, $x_i \sim N(0, 1)$, and various types of multicollinearity were introduced through predefined covariance matrices.

	Sample sizes		Numbers of features		β
	Training	Validation*	Total	Non-zeros	
Dataset I	50	10	100	10	$(3, 3, -3, 2, 2, -2, 1.5, 1.5, 1.5, -1.5, 0, \dots, 0)$
Dataset II	100	20	1,000	50	$\beta_1, \dots, \beta_{50} \sim N(0, 4)$, others zero
Dataset III	200	40	10,000	50	$\beta_1, \dots, \beta_{50} \sim N(0, 4)$, others zero
Dataset IV	400	80	10,000	50	$\beta_1, \dots, \beta_{50} \sim N(0, 4)$, others zero

Table 1. Description of the simulation data. The validation data is further generated with the size equivalent to 20% of the training data.

- Dataset I: 100 variables, 50 training samples (+10 validation), 10 non-zero coefficients, $\beta = (3, 3, -3, 2, 2, -2, 1.5, 1.5, 1.5, -1.5, 0, \dots, 0)$
- Dataset II: 1,000 variables, 100 training samples (+20 validation), 50 non-zero coefficients, $\beta_1, \dots, \beta_{50} \sim N(0, 4)$, others zero
- Dataset III: 10,000 variables, 200 training samples (+40 validation), 50 non-zero coefficients, $\beta_1, \dots, \beta_{50} \sim N(0, 4)$, others zero
- Dataset IV: 10,000 variables, 400 training samples (+80 validation), identical model to Dataset III but with the sample size doubled

For each dataset, block-structured covariance matrices were used to introduce different degrees of multicollinearity among predictors, following the structures defined in Baek et al. Specifically, predictors within each block share a fixed pairwise correlation, while predictors outside these blocks were generated as independent.

For each dataset, validation data equal to 20% of the training sample size was additionally generated and used exclusively for hyperparameter tuning; it was not used

to evaluate prediction error, since only feature selection performance was assessed and no separate test set was constructed.

3-2) Feature Selection Models

$$\hat{\beta}^{Lasso} = \arg \min_{\beta} \left[\frac{1}{n} \sum_{i=1}^n \left(y_i - \beta_0 - \sum_{j=1}^p \beta_j x_{ij} \right)^2 + \lambda \sum_{j=1}^p |\beta_j| \right]$$

Using the four simulated datasets described above, we applied four feature selection methods: LASSO, Elastic Net, Random LASSO, and Hi-LASSO. While Baek et al. (2026) recently introduced Stochastic LASSO, we restricted our scope to validating the foundational progression up to Hi-LASSO, reserving the newest model for future comparative work.

LASSO applies an L1 penalty to the regression coefficients, performing variable selection and coefficient estimation simultaneously by shrinking irrelevant coefficients to exactly zero. Elastic Net combines L1 and L2 penalties, allowing groups of correlated predictors to be selected together, which addresses a key limitation of LASSO under multicollinearity. Random LASSO is a bootstrap-based method that repeatedly draws random subsets of predictors, applies LASSO to each bootstrap sample, and aggregates variable importance scores across bootstrap samples to estimate final coefficients. Hi-LASSO extends Random LASSO by incorporating statistical importance measures into the bootstrap sampling probabilities, improving feature selection performance in extremely high-dimensional, low-sample-size (EHDLSS) settings.

For LASSO and Elastic Net, the regularization parameters were tuned by minimizing the prediction error on the validation data. For Random LASSO and Hi-LASSO, the number of predictors drawn per bootstrap sample (q) was set equal to the number of training samples (n), and the number of bootstrap samples (B) was set to $(p/q) \times 30$, ensuring that each predictor was included in the bootstrap sampling procedure 30 times on average.

3-3) Evaluation Metrics

Feature selection performance was evaluated using the F1-score and the Area Under the Precision-Recall Curve (AUCPR). Non-zero coefficients were defined as positive, and zero coefficients were defined as negative. Based on this definition, a confusion matrix was constructed as follows: True Positive (TP) when a model correctly identified a non-zero variable as non-zero, False Positive (FP) when a model incorrectly identified a zero variable as non-zero, False Negative (FN) when a model incorrectly identified a non-zero variable as zero, and True Negative (TN) when a model correctly identified a zero variable as zero.

The F1-score was calculated as $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$, where $\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$ and $\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$. The AUCPR was computed from the precision-recall curve obtained by evaluating each model across varying selection thresholds.

To ensure the reproducibility of the results, all experiments were repeated ten times on independently generated synthetic datasets, and the mean and standard deviation of each metric were reported.

3-4) Empirical Thresholding & Standardized Selection

To establish a robust and equitable baseline for comparison, our implementation isolated the core architectural differences between Random LASSO and Hi-LASSO. Traditional Random LASSO applications typically utilize arbitrary heuristic thresholds for final feature selection, while early Hi-LASSO implementations rely on a binomial hypothesis test. However, to account for the extreme variance and noise inherent in clinical genomic data—where rigid statistical cutoffs risk artificially truncating valid biological signals—the variable selection mechanism was standardized across both models. The native thresholding rules were bypassed in favor of an adaptive, relative-fractions threshold grid optimized directly against the validation RMSE.

Furthermore, while the original Hi-LASSO literature utilizes adaptive inverse weighting during the second bootstrap phase, this mechanism was also intentionally omitted to

reduce the risk of over-shrinkage of fragile biomarkers. This standardized approach allowed us to explicitly isolate two specific mathematical advancements of the internal Hi-LASSO architecture. First, during the initial importance scoring, highly correlated predictors with opposing signs can cause coefficient estimates to flip across different bootstrap samples. Random LASSO calculates importance by taking the absolute value of the averaged coefficients, which allows positive and negative estimates to mathematically cancel each other out. Hi-LASSO resolves this by averaging the absolute values of the coefficients, ensuring that opposing signs reinforce rather than diminish a feature's true importance score.

Second, Hi-LASSO structurally avoids the "dilution trap" that introduces severe systematic bias in traditional Random LASSO implementations. In both its importance scoring and final coefficient calculations, Random LASSO divides coefficient sums by the total number of bootstraps (B). This mathematically forces unselected predictors in any given iteration to be evaluated as exact zeros, artificially shrinking their final magnitude. In contrast, Hi-LASSO treats unselected iterations as missing values (NaN) rather than forced zeros. Therefore, the final coefficient estimates are normalized strictly by the exact number of times a feature is actively selected, mathematically preventing the artificial dilution of the coefficients.

3-5) Real Data Application

To evaluate the practical applicability of the four feature selection methods, the pipeline was applied to a real genomic dataset combining Glioblastoma Multiforme (GBM) and Brain Lower Grade Glioma (LGG) patients. The dataset consists of gene expression profiles (X) for 266 patients across 19,777 genes, along with the corresponding survival time (y) as the response variable. Each of the 19,777 predictors corresponds to a named gene.

Using the standardized Hi-LASSO pipeline configured with the empirically optimized thresholding strategy described above, and adjusting the hyperparameter search boundaries to accommodate the extreme sparsity of the clinical dataset, we selected the

top 10 genes ranked by the absolute value of their estimated coefficients. To assess their biological relevance, we searched the biomedical literature for prior evidence linking each gene to glioblastoma or glioma.

4. Experimental Results

4-1) Simulation Results

Figure 2 presents boxplots of F1-score (A) and AUCPR (B) for LASSO, Elastic Net, Random Lasso, and Hi-LASSO across the four simulated datasets (Dataset I–IV), based on 10 independent repetitions.

For F1-score, the two bootstrap-based methods, Random Lasso and Hi-LASSO, consistently outperformed the non-bootstrap-based LASSO and Elastic Net, and this gap widened as dimensionality increased. In Dataset I ($p=100$), all four methods showed relatively comparable performance, with Hi-LASSO achieving the highest median and the smallest variance. As the number of variables increased to 1,000 and 10,000 in Dataset II–IV, LASSO and Elastic Net showed substantially lower F1-scores with greater variability, while Random Lasso and Hi-LASSO remained relatively high and stable. Between the two bootstrap-based methods, Hi-LASSO generally achieved the highest or comparable median F1-score to Random Lasso, except in Dataset III, where Random Lasso slightly outperformed Hi-LASSO.

For AUCPR, the overall trend was similar to F1-score, though the gap between methods was narrower, particularly in the high-dimensional Dataset III. Hi-LASSO achieved the highest or near-highest median AUCPR across all four datasets, indicating that it ranks true non-zero variables reliably even without applying a hard selection threshold. LASSO consistently showed the lowest AUCPR, especially in Dataset III and IV, reflecting its known limitation in extremely high-dimensional, low-sample-size (EHDLS) settings.

Overall, these results indicate that bootstrap-based feature selection methods, particularly Hi-LASSO, are more robust to increasing dimensionality and multicollinearity than conventional L1- or L1+L2-penalized regression, consistent with findings reported in prior bootstrap-based LASSO studies.

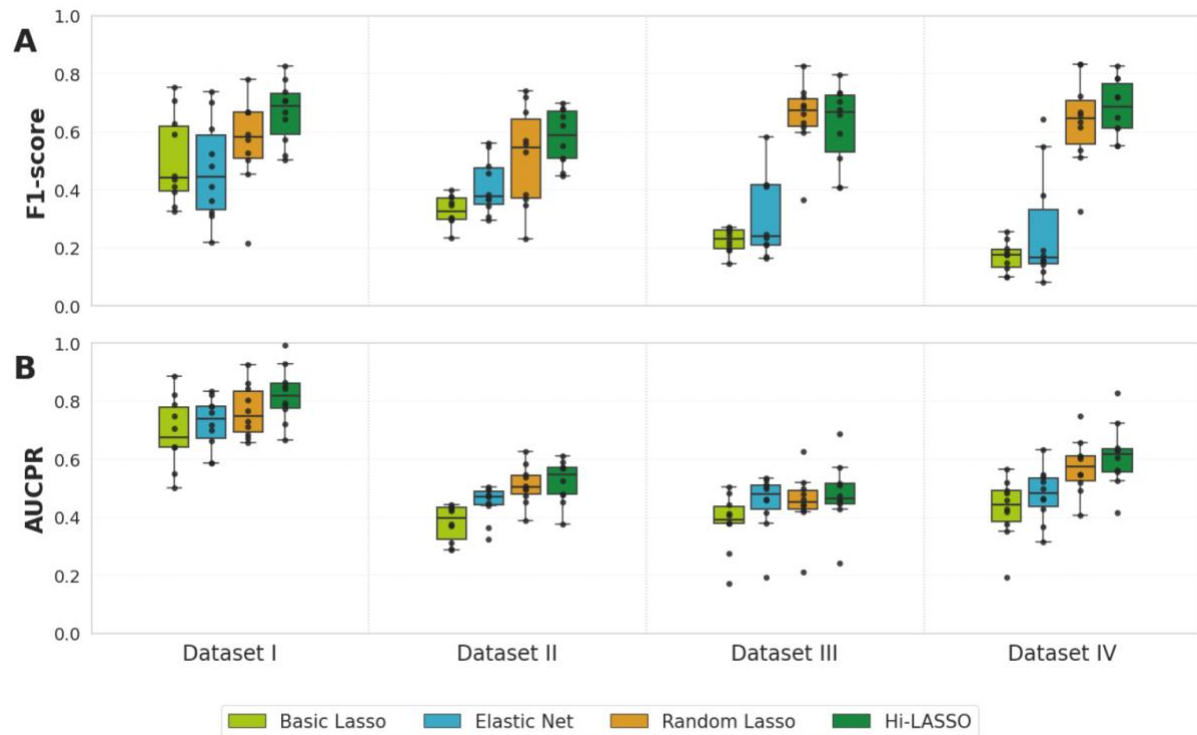


Figure 1. Feature selection performance with simulation data. (A) Comparison of F1-score. (B) Comparison of AUCPR. Boxplots show the distribution of each metric across 10 repeated experiments for LASSO, Elastic Net, Random Lasso, and Hi-LASSO on Dataset I-IV.

4-2) Real Data Results

Table 2 presents the top 10 genes identified by Hi-LASSO in the GBM and LGG dataset, ranked by the absolute value of their estimated coefficients, along with their Pearson correlation with survival time. Three of these genes have prior support in the biological literature. *ONECUT3* was reported to suppress glioblastoma cell proliferation and promote a glial-to-neuronal identity switch, implicating it in GBM reprogramming. *HILS1* was identified as a strong prognostic biomarker for LGG, with higher expression associated with higher tumor grade and worse survival; our analysis assigned *HILS1* a negative coefficient, consistent with this direction. *COL22A1* was reported to be overexpressed in GBM and identified as a key angiogenesis-related gene in high-grade glioma, with elevated expression associated with poor overall survival, aligning with its negative coefficient in our results. The remaining seven genes (*DEFB120*, *GSC2*, *UGT1A10*, *AMTN*, *LINC00442*, *USP17L6P*, and *GPR50*) were not found to have direct prior

associations with GBM or LGG in the literature reviewed, and are therefore proposed as candidate biomarkers warranting further biological validation.

Gene	Coefficient	Correlation	Reference
DEFB120	0.115154	0.501925	-
ONECUT3	0.104386	0.476075	5
GSC2	0.091427	0.450987	-
COL22A1	-0.082008	-0.302456	9-11
UGT1A10	0.070764	0.379763	-
AMTN	0.064322	0.452014	-
LINC00442	0.060331	0.449525	-
HILS1	-0.059956	-0.275192	6-8
USP17L6P	0.057047	0.375904	-
GPR50	0.056356	0.360708	-

Table 2. Top 10 ranked genes by Hi-LASSO in GBM and LGG.

5. References

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6. Contribution

Boseung Jung: Investigation (experimental setting), Analysis, Visualization (presentation materials) and Literature survey;

Seongyun Ryu: Conceptualization (Background and Significance), Investigation (experimental setting), Methodology, Analysis and Report Writing;

Minseo Lee: Conceptualization (Background and Significance), Analysis, Visualization (presentation materials), Literature survey;

Axel Botticelli: Conceptualization (Background and Significance), Investigation (experimental setting), Analysis, Visualization (results).